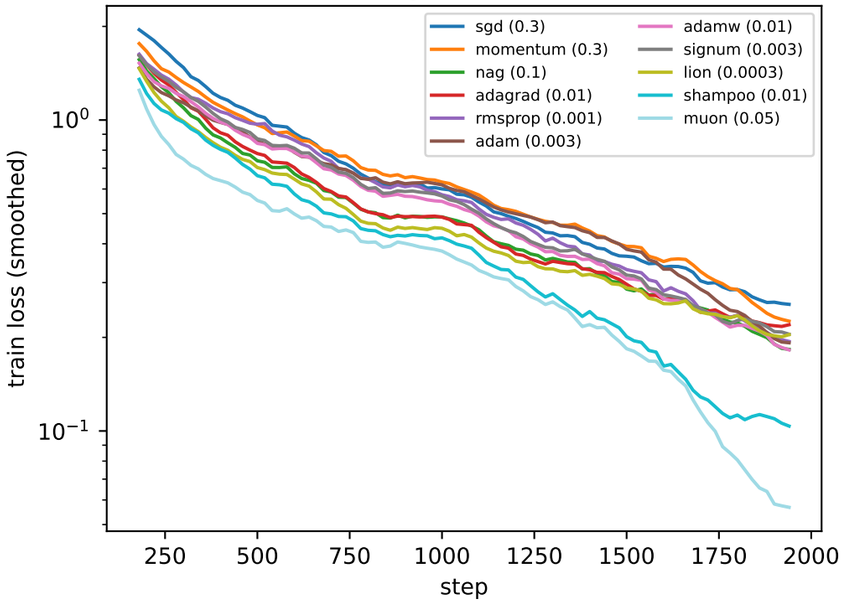
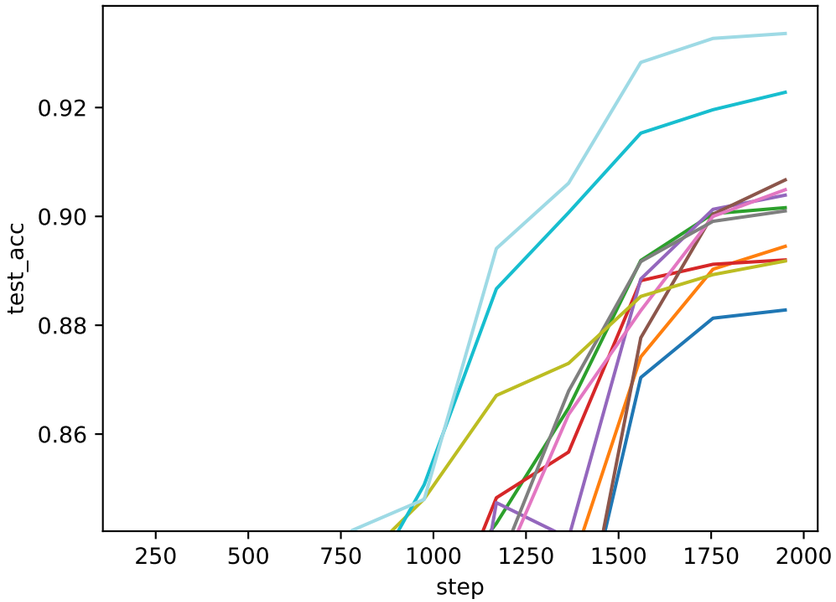
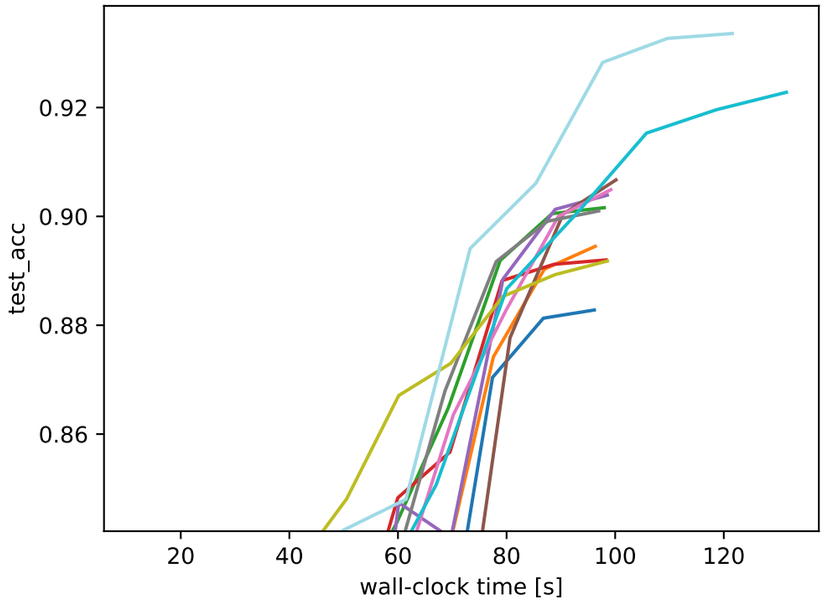

Training loss at best lr
train loss (smoothed)
$10^0$
$10^{-1}$
250
500
750
1000
1250
1500
1750
2000
step
sgd (0.3)
momentum (0.3)
nag (0.1)
adagrad (0.01)
rmsprop (0.001)
adam (0.003)
adamw (0.01)
signum (0.003)
lion (0.0003)
shampoo (0.01)
muon (0.05)


test_acc (mean ± std over seeds)
test_acc
0.92
0.90
0.88
0.86
250
500
750
1000
1250
1500
1750
2000
step


test_acc vs. time
test_acc
0.92
0.90
0.88
0.86
20
40
60
80
100
120
wall-clock time [s]